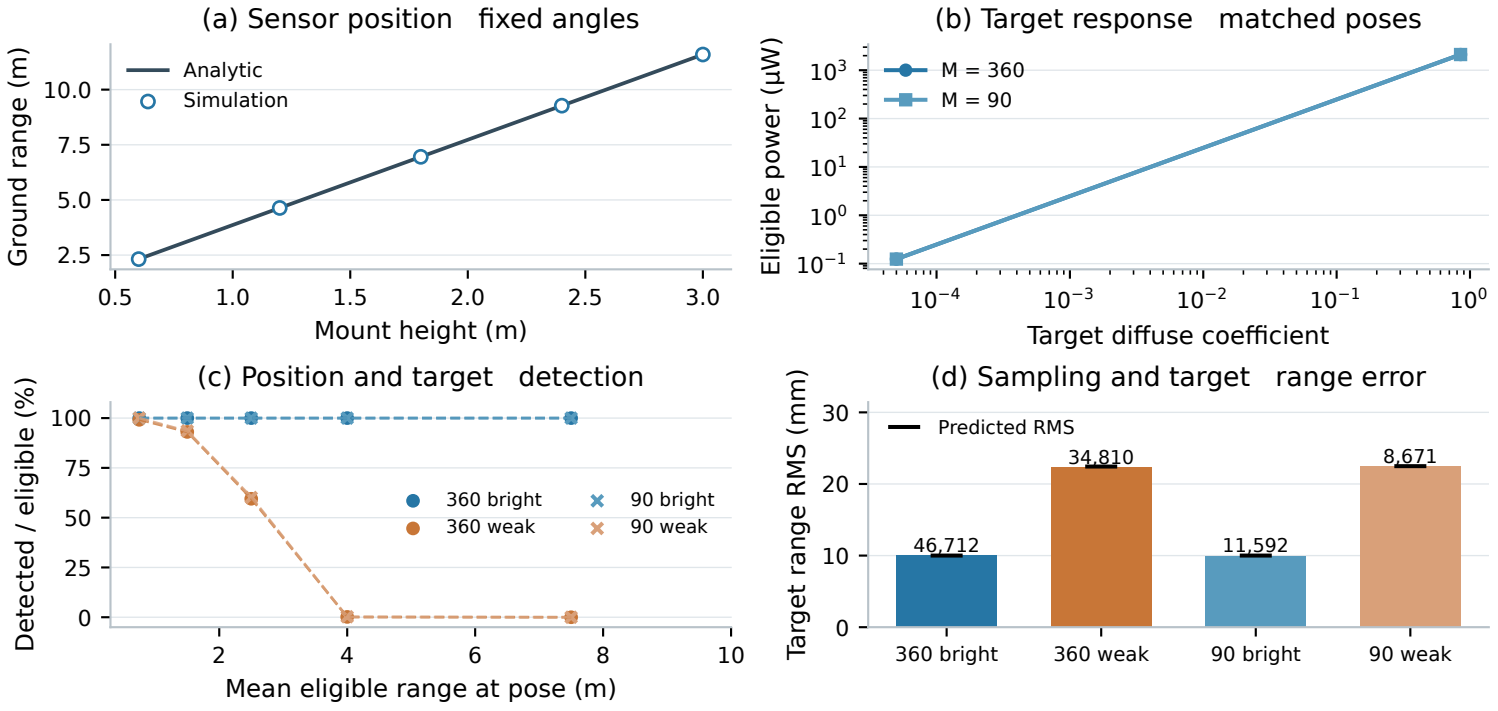
(a) Sensor position fixed angles
Analytic
Simulation
Ground range (m)
10.0
7.5
5.0
2.5
0.5
1.0
1.5
2.0
2.5
3.0
Mount height (m)
(b) Target response matched poses
M = 360
M = 90
Eligible power (μW)
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
Target diffuse coefficient
(c) Position and target detection
360 bright
360 weak
90 bright
90 weak
Detected / eligible (%)
100
75
50
25
0
2
4
6
8
10
Mean eligible range at pose (m)
(d) Sampling and target range error
Predicted RMS
Target range RMS (mm)
30
20
10
0
46,712
34,810
11,592
8,671
360 bright
360 weak
90 bright
90 weak